

Compiler vlastního programovacího jazyka

Ondřej Suk

13. května 2026

$$\frac{5 \cdot (\textit{fahrenheit} - 32)}{9}$$

$$5 * (\textit{fahrenheit} - 32) / 9$$

$$\boxed{5} \text{ } *(\textit{fahrenheit} - 32)/9$$

$\boxed{5} \boxed{*} (\textit{fahrenheit} - 32)/9$

5 * (*fahrenheit* - 32)/9

5 * (fahrenheit - 32) / 9

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Vstup

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Vstup

5 * (fahrenheit - 32) / 9

Vstup

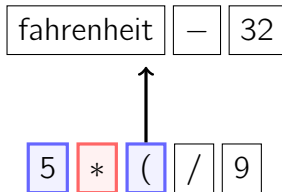
5 * (fahrenheit - 32) / 9

fahrenheit − 32

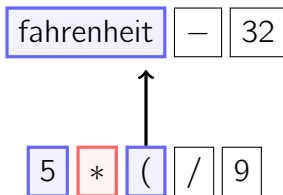
5 * (

/ 9

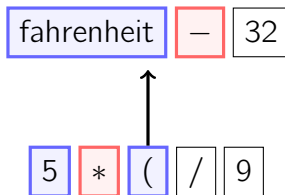
Vstup



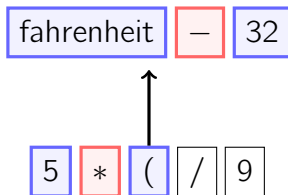
Vstup



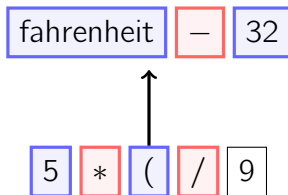
Vstup

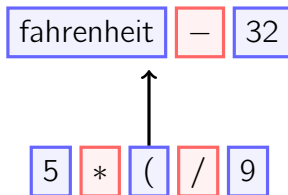


Vstup

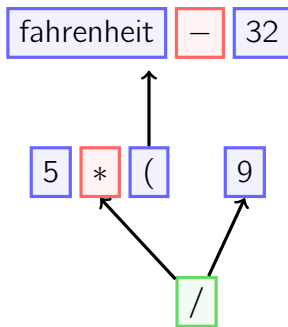


Vstup

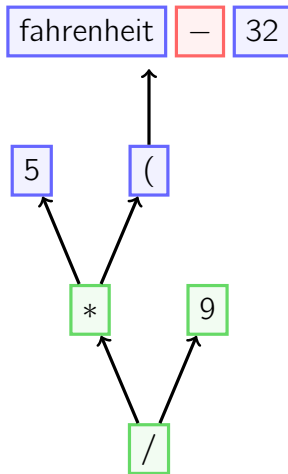




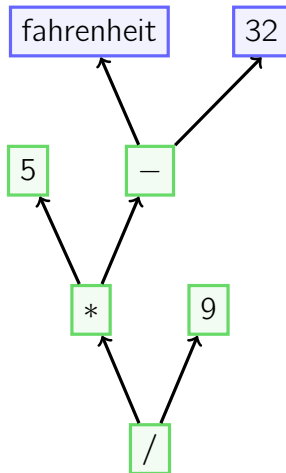
Vstup



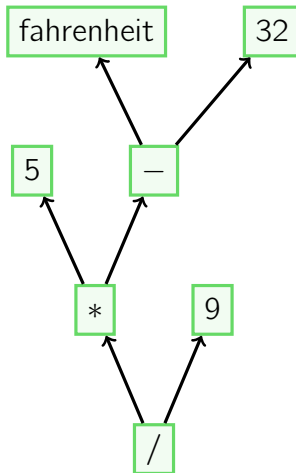
Vstup



Vstup

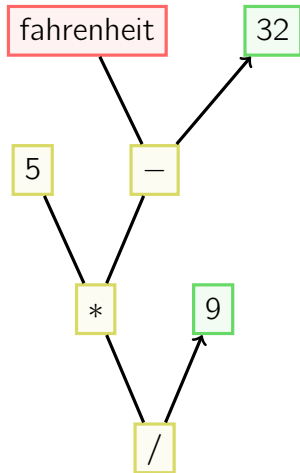


Vstup



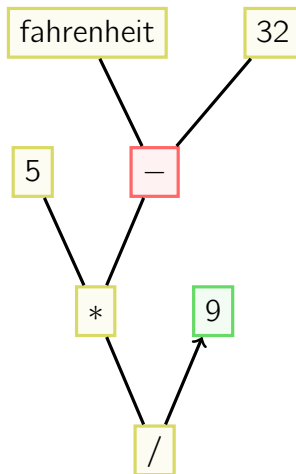
Vstup

```
%0 = load i64, ptr %fahrenheit, align 8
```



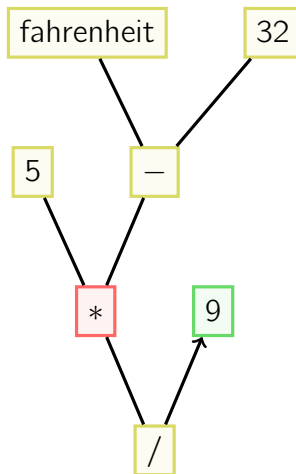
Vstup

```
%0 = load i64, ptr %fahrenheit, align 8  
%1 = sub i64 %0, 32
```



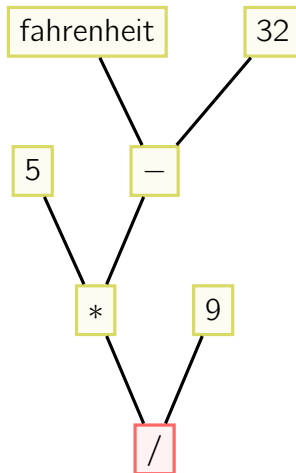
Vstup

```
%0 = load i64, ptr %fahrenheit, align 8  
%1 = sub i64 %0, 32  
%2 = mul i64 5, %1
```



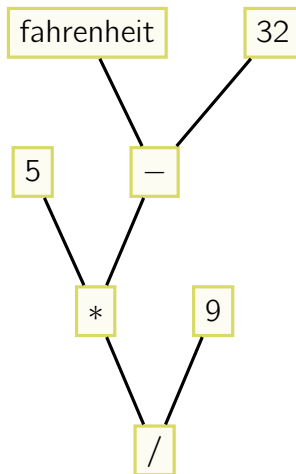
Vstup

```
%0 = load i64, ptr %fahrenheit, align 8  
%1 = sub i64 %0, 32  
%2 = mul i64 5, %1  
%3 = sdiv i64 %2, 9
```



Vstup

```
%0 = load i64, ptr %fahrenheit, align 8  
%1 = sub i64 %0, 32  
%2 = mul i64 5, %1  
%3 = sdiv i64 %2, 9
```



Vstup

```
%0 = load i64, ptr %fahrenheit, align 8
%1 = sub i64 %0, 32
%2 = mul i64 5, %1
%3 = sdiv i64 %2, 9
```

Vstup

```
leaq    (%rdi,%rdi,4), %rax
addq    $-160, %rax
movabsq $2049638230412172402, %rcx
imulq   %rcx
movq    %rdx, %rax
shrq    $63, %rax
addq    %rdx, %rax
```


Ukázka

```
~/Lang/cppcompiler
$ cat code
fn printnum(n) : {
    putchar 32;
    number n;
    putchar 10;
};

fn number(num) : {
    if (num >= 10) {
        number (num / 10);
    } else 0;
    putchar (48 + num % 10);
};

let a1 = 0;
let a2 = 1;

fn main() : {
    let i = 0;
    printnum 0;
    printnum 1;
    while (i < 64) {
        let a3 = a1 + a2;
        printnum a3;
        a1 = a2;
        a2 = a3;
        i = i + 1;
    };
};
```

```
$ ./program.exe
-bash: ./program.exe: No such file or directory
$ ./lang.exe code -o program.exe
$ ./program.exe
0
1
1
2
3
5
8
13
21
34
55
89
144
233
377
610
987
1597
2584
4181
6765
10946
17711
28657
46368
```

```
$ ./lang.exe code -h
```

```
USAGE:
```

```
lang.exe <vstup> [OPTIONS]
```

```
ARGUMENTS:
```

```
<vstup>          soubor ke kompilaci
```

```
OPTIONS:
```

```
-o <výstup>      Název zkompilevaného souboru (default: program.exe)
```

```
-a               Assembly výstup
```

```
-d               Debug výstup
```

```
-h               Seznam argumentů
```

```
$
```

Děkuji za pozornost!